

RAHW2

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## Question 1

(i)

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} \left[ xy (x^2 - y^2) (x^2 + y^2)^{-1} \right] . \\ &= (3x^2y - y^3) (x^2 + y^2)^{-1} + (x^3y - xy^3) (-1) (x^2 + y^2)^{-2} (2x)\end{aligned}$$

$$\frac{\partial f}{\partial x} = \frac{(yx^4 + 4x^2y^3 - y^5)}{(x^2 + y^2)^2}.$$

WRT y:

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} (x^3y - x^3) (x^2 + y^2)^{-1} . \\ &= (x^3 - 3y^2x) (x^2 + y^2)^{-1} + 2y (x^3y - xy^3) (-1) (x^2 + y^2)^{-2}\end{aligned}$$

$$\frac{\partial f}{\partial y} = \frac{x^5 - 4x^3y^2 - xy^4}{(x^2 + y^2)^2}.$$

Now we have at 0

$$\begin{aligned}\frac{\partial f}{\partial x}|_{0,0} &= \lim_{h \rightarrow 0} \frac{f(0 + h, 0) - f(0, 0)}{h} . \\ &= \lim_{h \rightarrow 0} \left( \frac{0 - 0}{h} \right)\end{aligned}$$

$$\frac{\partial f}{\partial x}|_{0,0} = 0.$$

And similarly

$$\begin{aligned}\frac{\partial f}{\partial y}|_{0,0} &= \lim_{h \rightarrow 0} \frac{f(0, 0 + h) - f(0, 0)}{h} . \\ &= \lim_{h \rightarrow 0} \frac{0 - 0}{h} .\end{aligned}$$

$$\frac{\partial f}{\partial y}|_{0,0} = \lim_{h \rightarrow 0} 0 = 0.$$

(ii)

$$\frac{\partial}{\partial x} \frac{\partial f}{\partial y} = \frac{\partial}{\partial x} \left( \frac{x^5 - 4x^3y^2 - xy^4}{(x^2 + y^2)^2} \right)$$

$$= (5x^4 - 12x^2y^2 - y^4) (x^2 + y^2)^{-2} + (-4x) (x^5 - 4x^3y^2 - xy^4) (x^2 + y^2)^{-3}.$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{5x^4 - 12x^2y^2 - y^4}{(x^2 + y^2)^2} - \frac{4x (x^5 - 4x^3y^2 - xy^4)}{(x^2 + y^2)^3}.$$

And for the other direction.

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial x} \left( \frac{yx^4 + 4x^2y^3 - y^5}{(x^2 + y^2)^2} \right).$$

$$= (x^4 + 12x^2y^2 - 5y^4) (x^2 + y^2)^{-2} + (yx^4 + 4x^2y^3 - y^5) (-4y) (x^2 + y^2)^{-3}.$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{(x^4 + 12x^2y^2 - 5y^4)}{(x^2 + y^2)^2} - \frac{4y (yx^4 + 4x^2y^3 - y^5)}{(x^2 + y^2)^3}.$$

Now we look at limit defn at zero.

$$\begin{aligned} \frac{\partial^2 f}{\partial y \partial x} \Big|_{0,0} &= \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial y} \Big|_{(0+h,0)} - \frac{\partial f}{\partial y} \Big|_{(0,0)}}{h}. \\ \frac{\partial^2 f}{\partial y \partial x} \Big|_{0,0} &= \lim_{h \rightarrow 0} \frac{\left( \frac{(0+h)^5 - 4(0+h)^3(0) - (0+h)(0)^4}{((0+h)^2 + (0)^2)^2} \right) - 0}{h}. \\ \frac{\partial^2 f}{\partial y \partial x} \Big|_{0,0} &= \lim_{h \rightarrow 0} \frac{0}{h} = \lim_{h \rightarrow 0} 0 = 0. \end{aligned}$$

In the other order we have:

$$\begin{aligned} \frac{\partial^2 f}{\partial x \partial y} \Big|_{0,0} &= \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial x} \Big|_{(0,0+h)} - \frac{\partial f}{\partial x} \Big|_{(0,0)}}{h}. \\ \frac{\partial^2 f}{\partial x \partial y} \Big|_{0,0} &= \lim_{h \rightarrow 0} \frac{\left( \frac{(0)^5 - 4(0)^3(0+h) - (0)(0+h)^4}{(0)^2 + (0+h)^2} \right) - 0}{h}. \\ \frac{\partial^2 f}{\partial x \partial y} \Big|_{0,0} &= \lim_{h \rightarrow 0} \frac{0 - 0}{h} = \lim_{h \rightarrow 0} 0 = 0. \end{aligned}$$

## Question 2

(i)

We are given:

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

We have a function  $f(x, y)$  which can be reparameterised under the reparametrisation  $x = \cos(t)$  and  $y = \sin(t)$ . Thus the total derivative is:

$$\begin{aligned} \frac{df}{dx} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}. \\ \frac{df}{dx} &= \frac{\partial f}{\partial x} (-\sin(t) dt) + \frac{\partial f}{\partial y} (\cos(t) dt) \end{aligned}$$

At an extremum we have

$$\frac{df}{dx} = \left( \frac{\partial f}{\partial x} \right) \Big|_{x_0, y_0} (-\sin(t_0)) + \frac{\partial f}{\partial y} \Big|_{x_0, y_0} (\cos(t_0)) = 0$$

We have  $x_0 = \cos t_0$  and  $y_0 = \sin t_0$

$$\frac{df}{dx} = - \left( \frac{\partial f}{\partial x} \right) \Big|_{x_0, y_0} y_0 + \frac{\partial f}{\partial y} \Big|_{x_0, y_0} x_0 = 0.$$

$$\left( \frac{\partial f}{\partial x} \right) \Big|_{x_0, y_0} y_0 = \frac{\partial f}{\partial y} \Big|_{x_0, y_0} x_0.$$

We know  $x_0, y_0$  are non-zero so we can pick a  $\lambda \in \mathbb{R}$  s.t

$$\lambda = \frac{\left( \frac{\partial f}{\partial x} \right) \Big|_{x_0, y_0}}{2x_0} = \frac{\left( \frac{\partial f}{\partial y} \right) \Big|_{x_0, y_0}}{2y_0}.$$

Thus we have show  $\exists \lambda \in \mathbb{R}$  s.t

$$2x_0\lambda = \left( \frac{\partial f}{\partial x} \right) \Big|_{x_0, y_0}.$$

$$2y_0\lambda = \left( \frac{\partial f}{\partial y} \right) \Big|_{x_0, y_0}.$$

we know  $\lambda \in \mathbb{R}$  as  $x_0, y_0$  and both partials are real.

(ii)

Extrema of  $f(x, y) = x^3 + y^3$  in  $S^1$ .

$$\frac{\partial f}{\partial x} = 3x^2 = 0.$$

$$\frac{\partial f}{\partial y} = 3y^2 = 0.$$

We can ignore the  $x_0 = 0$  and  $y_0 = 0$  cases as question states "using this". The values of  $X = 0$  and  $Y = 0$  satisfy the above PDEs and thus are maxima, but in part (i) we assumed both  $x$  and  $y$  nonzero.

Thus using the required method we have:

$$3x_0^2 = 2x_0\lambda.$$

$$3y_0^2 = 2y_0\lambda.$$

If we assume  $x_0$  and  $y_0$  are non zero we get

$$\lambda = \frac{3x_0}{2}.$$

and

$$\lambda = \frac{3y_0}{2}.$$

so  $x_0^2 = y_0^2$  and we have  $x_0^2 + y_0^2 = 1$  so  $x_0 = y_0 = \pm \frac{\sqrt{2}}{2}$ .

Thus  $\exists$  four extrema in  $S^1$  given by  $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

We have  $\lambda = \frac{3x_0}{2}$  or  $\lambda = \frac{3y_0}{2}$ . Thus  $\lambda = \frac{3\sqrt{2}}{2}$  for the positive, positive extrema and  $\lambda = -\frac{3\sqrt{2}}{2}$  for the negative, negative term. While  $\lambda$  isn't defined for the two points with alternating signs as  $x_0 \neq y_0$  so they don't satisfy both equations for  $\lambda$ .